

Hence the answer is A.

8

B

Total work done by man = Work done against friction + gain in GPE (since KE is constant)

$$\begin{aligned}\text{Work done against friction} &= 1500 - mgh \\ &= 1500 - 5.0 \times 9.81 \times 12 \\ &= 911.4 \text{ J}\end{aligned}$$

Average friction = work done against friction / distance travelled along plane

$$= 911.4 / \left(\frac{12}{\sin 30^\circ} \right) = 37.9 \approx 38 \text{ N}$$

9

B